Boules A. Marzouk

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Education

Ain Shams University *B.Sc. in Computer Science; GPA: 3.6 / 4.0* Cairo, Egypt

Experience

Software Engineer Feb 2026 – Present

Procore *Remote*

- Engineered and scaled a mission-critical internal platform using **Golang** and **Ruby on Rails**, streamlining workflows and integrating with **50+ systems** for over **30 engineering teams**.
- Served on a global **on-call rotation** to monitor enterprise CI/CD pipelines, and rapidly resolve production incidents, maintaining reliability.
- Engineered a cross-platform Node.js plugin for Claude Code and Cursor that **reduced main-context token consumption by 46–99.8%** across benchmarked tasks by intercepting and redirecting large whole-file reads.
- Optimized a service CI pipeline by sharing repeated tasks on a common shareable builder which **improved the CI time by 50%** cutting it down **from 10 minutes to 5 minutes** which was beneficial for quick iterations especially when working on multiple tasks with agents.
- Hardened execution reliability by designing fail-open pre-read hooks to prevent workflow blocking, resolving CodeQL security vulnerabilities (TOCTOU file-system races and shell injections), and **validating cross-environment behavior with 66 automated tests**.

Software Engineer Feb 2025 – Dec 2025

Siemens Digital Industries Software *Hybrid, Cairo*

- Refactored a critical backend API, reducing execution time by **80%** and improving overall system performance.
- Collaborated in **on-call rotations** with clients from 2+ major enterprise accounts, diagnosing and resolving production issues in real time, **resolving 90% of incidents within SLA**.
- Contributed to **gRPC protocol design**, enabling communication between **10+ internal services** and improving system interoperability.
- Developed a debugging capability still used by **40+ engineers**, reducing issue resolution time by **30%**.

Projects

Operating System Memory Manager | *C, OS Concepts, Concurrency* | GitHub Aug 2024

- Designed and implemented a custom **memory management system** simulating core OS functionalities, including **malloc, kernel-level malloc, and shared memory allocation**.
- Built **trap and fault handlers** to mimic real OS interrupt handling for memory errors.
- Developed a **process scheduler** from scratch with **concurrency control** using semaphores and locks.

Technical Skills

Languages: Java, GO, Ruby, C/C++, SQL, JavaScript, TypeScript
Backend: Node.js, Gin, NestJS, gRPC, REST APIs, Microservices
Messaging/Queues: Kafka, AWS SNS/SQS
Frontend: React, Stencil, Qt
Databases: PostgreSQL, MongoDB, MySQL
Tools: Git, Docker, AWS, CI/CD, Linux, VS Code
Other: Algorithms & Data Structures (400+ solved on LeetCode, Codeforces)

Volunteer

Vice-Chairman & Technical Team Head Oct 2022 – Sep 2026

ACMASCIS *Cairo, Egypt*

- Led the organization of **3+ major events**, coordinated **100+ active members** throughout the season.
- Contributed to multiple **organizational projects** that streamlined operations and supported ACM's mission.